HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

F78PB

Probability and Statistics B

Mid-Semester Test

Semester 2 2018/19

Duration: 50 minutes

IMPORTANT INFORMATION

There are six questions worth a total of 20 marks. Five are multiple choice and one requires a written answer. To achieve the full 20 marks you must answer all questions. Please answer the questions on the special sheet provided.

Note: This is a CLOSED BOOK test which lasts for 50 minutes. You may use a calculator and consult the distributions formula sheet which is provided.

1. Five students were asked how many pieces of fruit they had eaten one day. Their answers are as follows:

4, 3, 5, 2, 1

The sample standard deviation of these data is

(A) 3.25 (B) 3.11 (C) 2.17 (D) 1.64 (E) 1.58

[2 marks]

2. A study was carried out to investigate the salt content in bread sold in a particular large store. A sample of 30 different types of loaf was taken and the following results obtained.

1.2 6.3 8.0 9.1 10.5 11.8 12.0 12.2 13.6 13.8 14.3 14.7 15.7 15.9 16.5
16.8 17.8 19.0 19.7 20.8 21.8 23.2 23.6 27.0 28.7 29.0 32.1 33.8 40.9 42.1

Note that the results are listed in increasing salt content (measured in g per kg of bread).

A boxplot in R is obtained of these data. The results that are highlighted as outliers are:

(A) 1.2 only (B) 42.1 only (C) 1.2 and 42.1 (D) 40.9 and 42.1 (E) no outliers.

(Hint: The default boxplot in R has whiskers extending to 1.5 times the interquartile range).

[3 marks]

3. Suppose that $\underline{X} = (X_1, X_2, ..., X_n)$ is a random sample from a distribution with density function

$$f_X(x;\theta) = \begin{cases} \dfrac{3x^2}{\theta^3} & \text{if } 0 \le x \le \theta \\ 0 & \text{if } x < 0 \text{ or } x > \theta \end{cases}$$

The bias of the estimator $\hat{\theta} = \overline{X}$ is given by:

(A) $\frac{5}{3}\theta$ (B) $\frac{5}{3n}\theta^2$ (C) $\left(\frac{1}{16} + \frac{3}{80n}\right)\theta^2$ (D) $\frac{13}{80}\theta^2$ (E) $-\frac{1}{4}\theta$

[4 marks]

4. Let $\underline{X} = (X_1, X_2, \ldots, X_n)$ be a random sample of size n from a Gamma $(2, \frac{1}{\theta})$ distribution. This has mean 2θ and variance $2\theta^2$. Consider $Y = c\overline{X}$ as an estimator of θ where $c > 0$ is a constant. By using the basic results for the expected value and variance of $\overline{X}$, the MSE of Y is

(A) $\frac{2c^2\theta^2}{n} + (2c-1)^2\theta^2$ (B) $\frac{2c\theta^2}{n} + (c-1)\ \theta$ (C) $\frac{4c^4\theta^2}{n^2} + (2c-1)\ \theta^2$

(D) $\frac{2c^2\theta^2}{n} - (2c-1)\ \theta$ (E) $\frac{2c^2\theta^2}{3} + (2c-1)^2$

[4 marks]

5. In a small medical study, a random sample of 8 patients is followed up and the time T (in months) until they show a symptom is recorded. Observation starts for all patients at the same time and the study ends after 2 months.

At the end of the study period, 6 patents have not shown the symptom. The other 2 patients developed the symptom at 1.4 and 1.6 months. Given that $T \sim \text{Exp}(\lambda)$, the likelihood function is:

(A) $\lambda^2 e^{-15\lambda}$ (B) $\lambda^{10} e^{-5\lambda}$ (C) $10\lambda^4 e^{-60\lambda}$ (D) $3e^{-3\lambda}$ (E) $1.4\lambda + 1.6e^{-10\lambda}$

[4 marks]

6. Write the answer to this question on the special answer sheet in the space indicated there.

For an independent and identically distributed random sample of size n from the Beta(θ, 4) distribution find the method of moments estimate (MME) of θ.

[3 marks]

Some Absolutely Continuous Distributions

Name	Notation	p.d.f.	range	m.g.f.	mean	variance
Uniform	$U(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{b\theta}-e^{a\theta}}{(b-a)\theta}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu\theta+\frac{1}{2}\sigma^2\theta^2}$	μ	σ^2
Gamma	$\Gamma(t,\lambda)$	$\frac{\lambda^t}{\Gamma(t)}x^{t-1}e^{-\lambda x}$	$0<x<\infty$, $t,\lambda>0$	$\left(\frac{\lambda}{\lambda-\theta}\right)^t$	$\frac{t}{\lambda}$	$\frac{t}{\lambda^2}$
Exponential	$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-\theta}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_n	$\frac{1}{2^{n/2}\Gamma(n/2)}x^{n/2-1}e^{-x/2}$	$0<x<\infty$, $n>0$ integer	$(1-2\theta)^{-n/2}$	n	$2n$
Beta	$\beta(\alpha_1,\alpha_2)$	$\frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)}x^{\alpha_1-1}(1-x)^{\alpha_2-1}$	$0<x<1$, $\alpha_1,\alpha_2>0$	not useful	$\frac{\alpha_1}{\alpha_1+\alpha_2}$	$\frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)}$
Cauchy	–	$\frac{\lambda}{\pi(\lambda^2+(x-\mu)^2)}$	$-\infty<x<\infty$, $\lambda>0$	doesn't exist but char. fn. is $e^{i\mu\theta-\lambda\lvert\theta\rvert}$	no moments exist	
Multivariate Normal	$N(\mu,\Sigma)$	$\frac{\exp\left\{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right\}}{(2\pi)^{d/2}\det(\Sigma)^{1/2}}$	$x \in \mathbb{R}^d$	$e^{\mu^T\theta+\frac{1}{2}\theta^T\Sigma\theta}$	μ	covariance matrix Σ

Some Discrete Distributions

(All distributions below have integer-valued ranges)

Name	Notation	prob. fn.	range	m.g.f.	mean	variance
Binomial	$\text{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$0\le x\le n$ $0<p<1$	$[(1-p)+pe^\theta]^n$	np	$np(1-p)$
Poisson	$\text{Pois}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$0\le x<\infty$ $\lambda>0$ real	$e^{\lambda(e^\theta-1)}$	λ	λ
Hyper-geometric	$\text{Hgeo}(n,a,b)$	$\frac{\binom{a}{x}\binom{b}{n-x}}{\binom{a+b}{n}}$	$0\le x\le n$ $a,b>0$ integers	not useful	$\frac{na}{a+b}$	$\frac{nab(a+b-n)}{(a+b)^2(a+b-1)}$
Negative binomial	$\text{Nbin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$k\le x<\infty$, $0<p<1$	$p^k[e^{-\theta}-(1-p)]^{-k}$	k/p	$k(1-p)/p^2$
Geometric	$\text{Geo}(p)$	$p(1-p)^{x-1}$	$1\le x<\infty$ $0<p<1$	$p[e^{-\theta}-(1-p)]^{-1}$	$1/p$	$(1-p)/p^2$

Multinomial: multivariate extension of binomial

Prob. fn. is given by

$$f(x_1,x_2,\ldots,x_k) = \frac{n!}{x_!!x_2!\ldots x_k!x_{k+1}!}p_1^{x_1}p_2^{x_2}\ldots p_k^{x_k}p_{k+1}^{x_{k+1}}, \quad 0\le x_i\le n, \quad 0<p_i<1 \quad \forall i=1,2,\ldots,n+1$$

where $p_1+p_2+\cdots+p_{k+1}=1$ and $x_1+x_2+\cdots+x_{k+1}=n$.

The m.g.f. is given by

$$[p_1e^{\theta_1}+p_2e^{\theta_2}+\cdots+p_ke^{\theta_k}+p_{k+1}]^n.$$

Moments: $E(X_i)=np_i$, $\text{Var}(X_i)=np_i(1-p_i)$, $\text{cov}(X_i,X_j)=-np_ip_j$, $i,j=1,2,\ldots,n+1$.